

Minimum Viable Product (MVP) Definition and Rationale

Ecosystem Discovery Application

1. Introduction

The Minimum Viable Product (MVP) represents the smallest functional version of the Ecosystem Discovery Application that delivers meaningful educational value to users.

The purpose of defining an MVP is to identify the core system features required to demonstrate the main learning interaction of the application, while keeping the development scope manageable for the project timeline.

For this project, the MVP focuses on allowing students to explore a freshwater ecosystem, discover animals through interaction, and learn basic ecological information in a simple and engaging way.

2. MVP Vision

The envisioned MVP for the Ecosystem Discovery Application provides a single interactive freshwater ecosystem environment where students can explore and discover animals.

When users enter the ecosystem scene, they will see a visual environment representing a Saskatchewan-inspired freshwater habitat. Animals will appear within their ecological zones such as underwater, near vegetation, or along the shoreline.

Users can interact with the ecosystem by clicking on animals. When an animal is selected, the system displays educational information about that species and unlocks a sticker within a digital sticker book.

This discovery-based interaction encourages curiosity and supports the Grade 3 Habitats and Communities curriculum.

3. Core MVP Features

The MVP includes the following essential features:

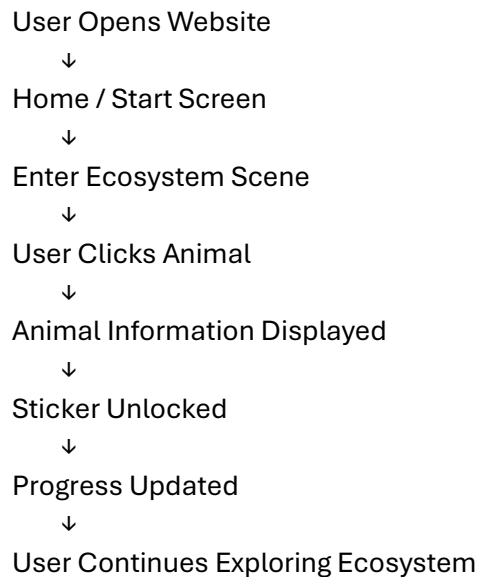
Feature	Description
Interactive Ecosystem Scene	A visual freshwater ecosystem representing a Saskatchewan habitat
Clickable Animal Discovery	Users can click animals within the ecosystem to discover them
Animal Information Display	Educational facts appear when an animal is discovered
Sticker Book System	Each discovered animal unlocks a collectible sticker

Feature	Description
Progress Tracking	Displays the number of animals discovered by the user
Animal Data Model	Animal information stored using JSON or lightweight storage
MVC Architecture	System structured using Model–View–Controller design

These features provide the minimum functionality required to demonstrate the educational experience of the application.

4. MVP Interaction Flow Diagram

The diagram below illustrates the core interaction flow of the MVP system.



This flow represents the core discovery interaction of the application.

5. MVP User Experience Flow

The MVP interaction flow is designed to be simple and intuitive for young learners.

1. The user opens the Ecosystem Discovery Application.
2. The home screen introduces the ecosystem learning experience.
3. The user enters the ecosystem scene.
4. The ecosystem environment displays animals within their natural habitat zones.
5. The user clicks on an animal.

6. The system displays educational information about the selected animal.
7. The corresponding sticker is unlocked in the sticker book.
8. The discovery progress tracker updates.
9. The user continues exploring the ecosystem to discover additional animals.

This interaction loop represents the core learning activity of the application.

6. MVP Rationale

The selected MVP features represent the minimum functionality required to deliver the educational objective of the system.

These features allow students to:

- Explore an ecosystem environment
- Discover animals through interaction
- Learn ecological information
- Receive visual rewards through the sticker collection system

By focusing on these essential interactions, the MVP demonstrates the core concept of the application while avoiding unnecessary complexity during early development.

This approach allows the development team to build a working and testable version of the system quickly, while leaving room for future improvements.